

Deep Learning 2

Special blocks

Machine learning

Roman Jakubíček

Department of Biomedical Engineering

Brno University of Technology

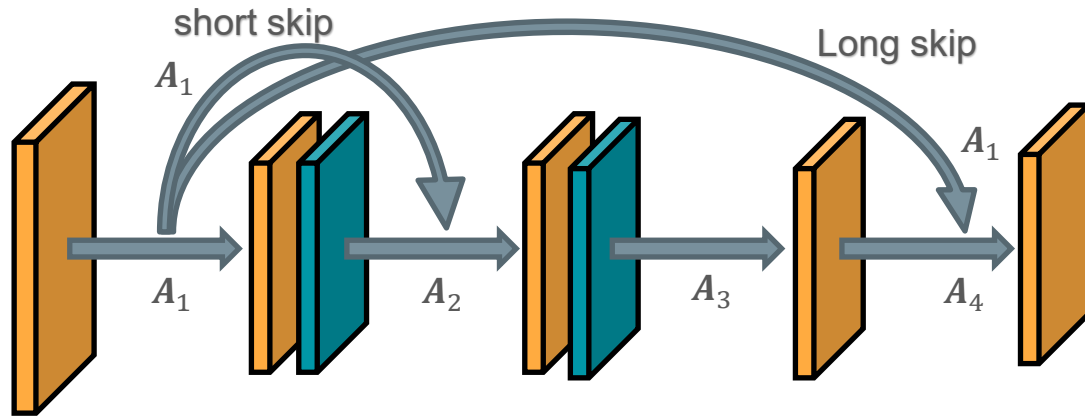
Introduction

- Other blocks and structures
 - Skip connections
 - 1×1 convolution, Bottleneck
 - Transposed and dilated convolution
 - Un-pooling layer
 - Residual block
 - Attention module

Special blocks

Skip connections

- Short or long skip connections
- Skip connections help traverse information in deep neural networks
- It is trying to find and to use the layers which are rich in features
- It provides improving the information flow between layers
- Solve vanishing or exploding problems (BatchNorm helps in depth) in "shallow" nets



... but, how do we can join data?

Output from l^{th} layer - A_l

Skip connections

Concatenation

- in the context of programming, is the operation of joining two strings together
- joining several same sized (can be ensured) activation maps into depth as additional features
- output tensor will be expanded in depth

$$A_l^* = [A_1, \dots, A_{l-1}] \quad \text{e.g.} \quad A_4^* = [A_1, A_4]$$

Additive (residual)

- add a skip-connection that bypasses the non-linear transformations with an identity function
- to adding, the tensors must be of the same-sized (it can be solved padding, interpolation or pooling)
- output will be same sized as input in depth

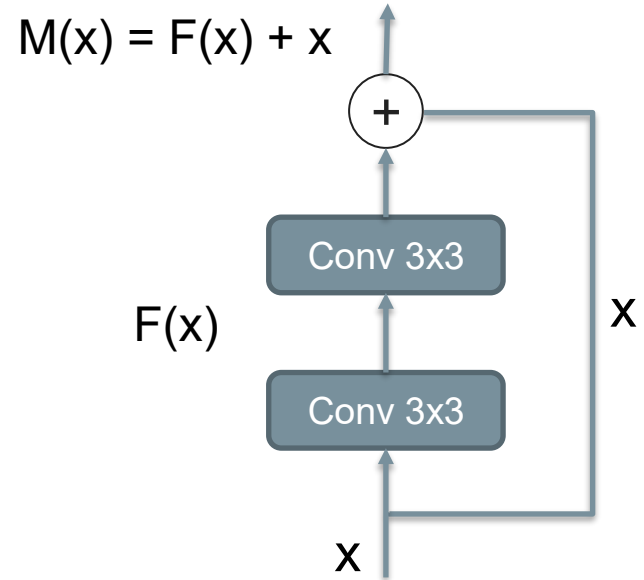
$$A_l^* = [A_1 + A_2 + A_{\{l-1\}}] \quad \text{e.g.} \quad A_4^* = [A_1 + A_4]$$

Residual block

- The deeper model should be able to perform at least as well as the shallower model
- but deeper models are harder to optimize
- It has two branches flowing into additive connection (identity and convoluted identity)
- Thus, the weights of conv layers are adapted this way, that the output from them will “residuum” defined as follows

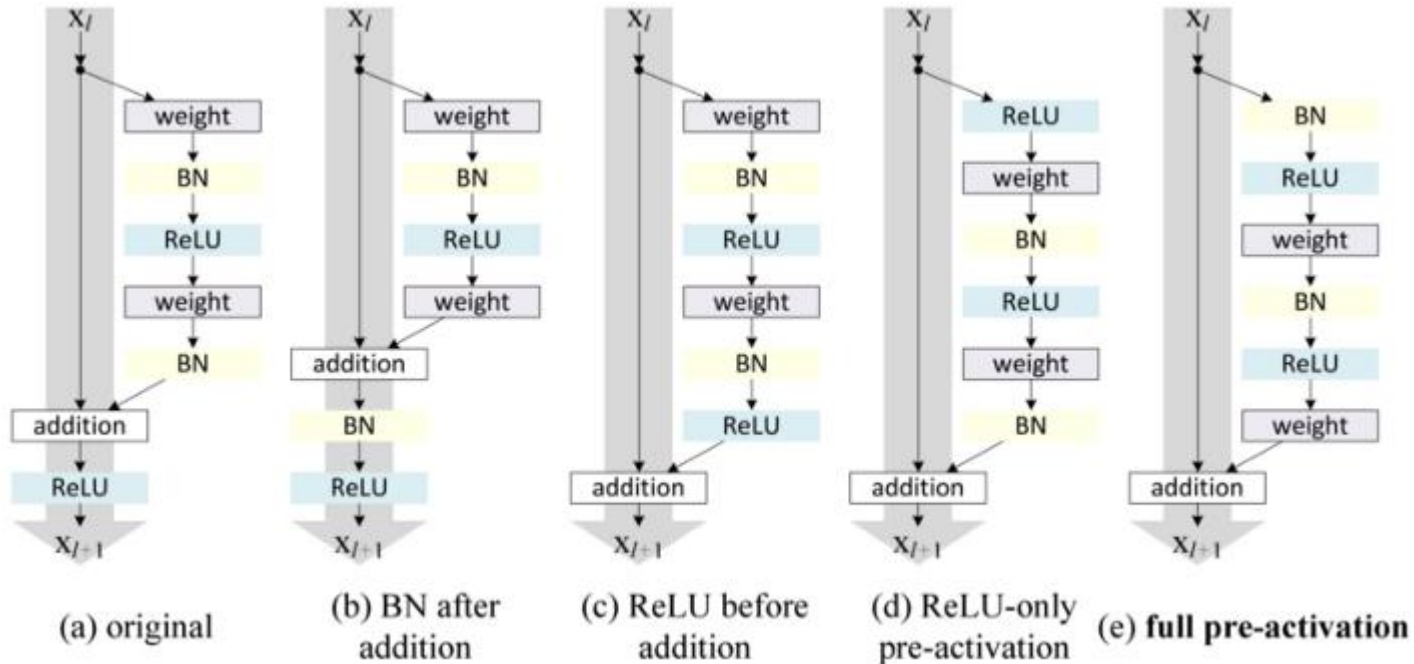
$$F(x) = M(x) - x$$

- In other words, this block can take the output from a series of conv layers or just original input (skip layer)
- Avoids vanishing gradients
- May be able to learn simple function (identity function)



Residual block

- There are many variations of block structure
- Can also be included 1x1 convolution in the identity branch

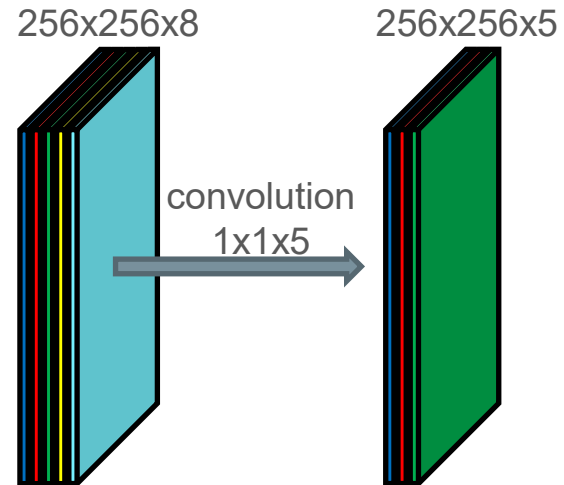


1 x 1 convolution layer

- Sometimes called “point-wise convolution”
- It is not combining spatial information, but it is combining features in depth
- The output is the same size as input in spatial domain, but the depth (number of features) can be changed.
- We obtain new activation maps (feature maps), which are a linear combination of previous features
- It has no hyper-parameters, but there are learned the weights of individual features

Why would we want 1x1 convolution?

- New combination features
- Reduce or expand features
- Speed up by decreasing the dimension
- Separable convolution (MobileNet)
- Alternative computation of FC layer (assuming $1 \times 1 \times N$ tensor as input)



Bottleneck – autoencoders

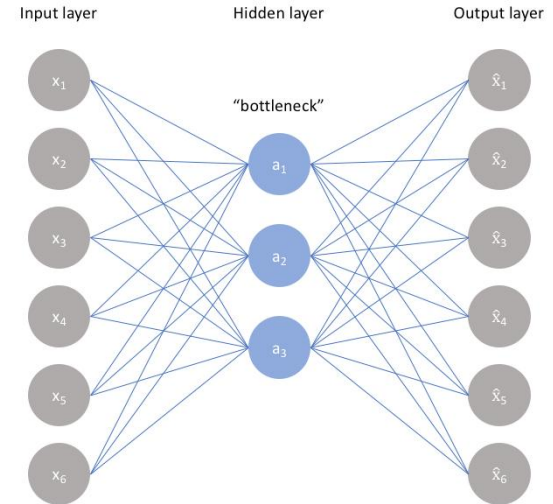
- constrains the amount of information that can traverse the full network, forcing a learned compression of the input data (“regularization”)
- It is designed for finding the best tradeoff between accuracy and complexity (compression)
- To find (learn) weights of A(D)NN for which it applies that

$$\min(||x - \hat{x}||)$$

- The bottleneck is a key property/part of the network architecture, without this bottleneck, our network could easily learn to simply memorize the input values by passing these values through the network.

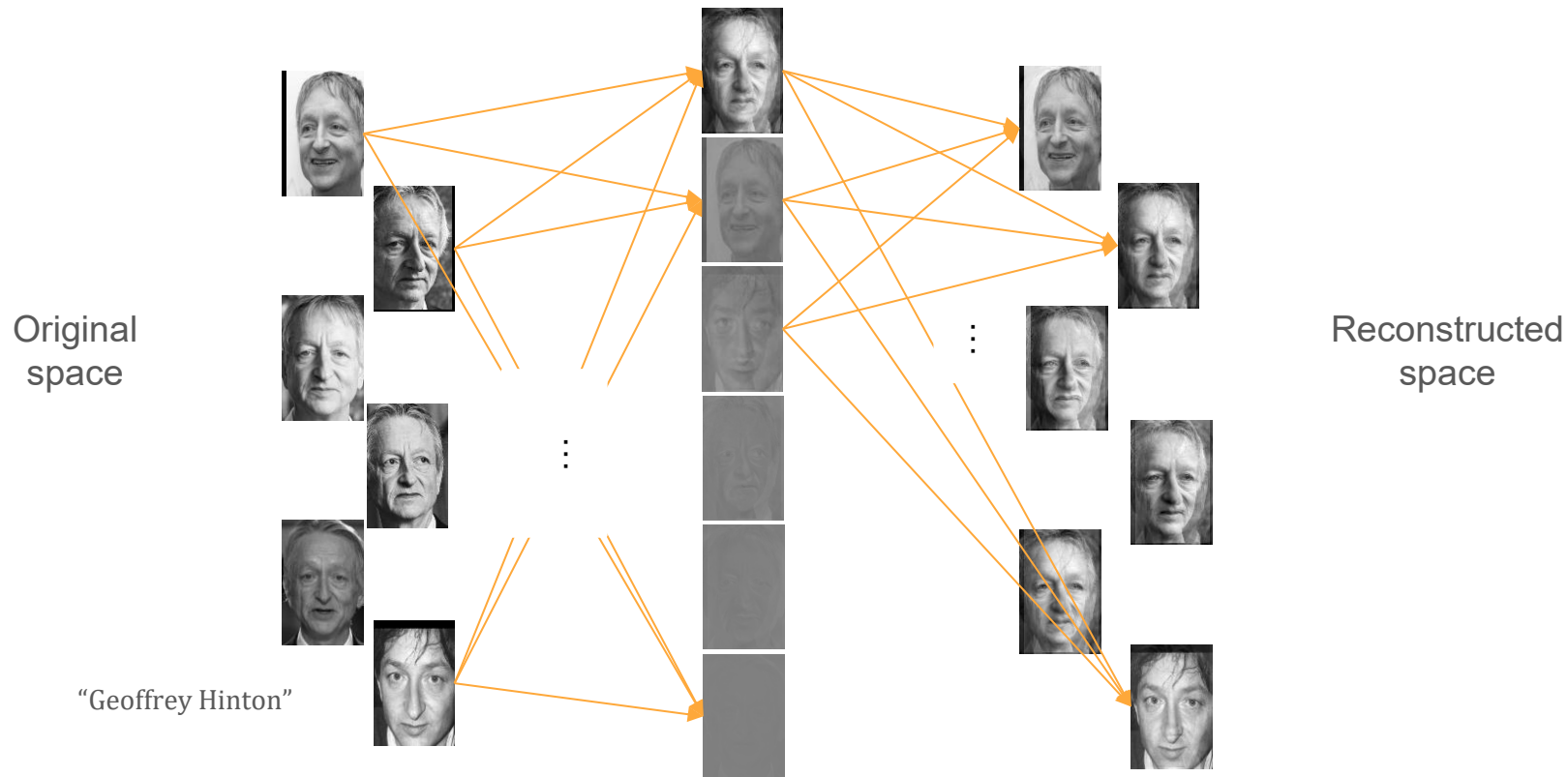
Applications:

- Dimensional reduction (compression)
- Encoder – decoder (U-net)
- Bottleneck layer (GoogLeNet)



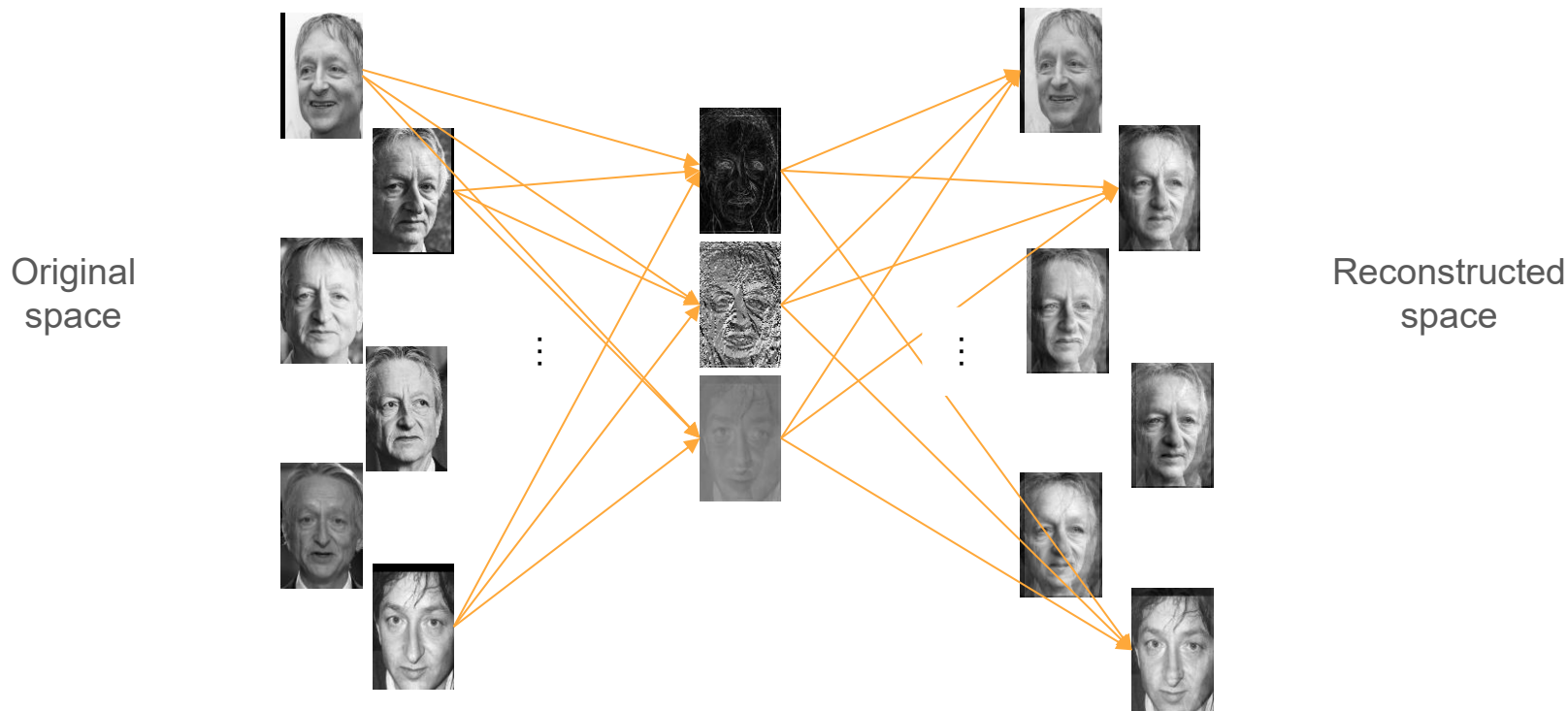
Bottleneck

Example feature reduction for face recognition by PCA



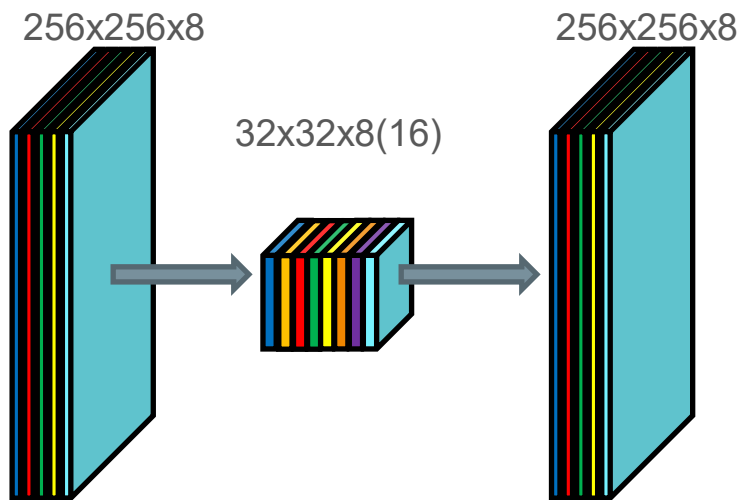
BottleNeck

Example feature (information) reduction for face recognition by NN

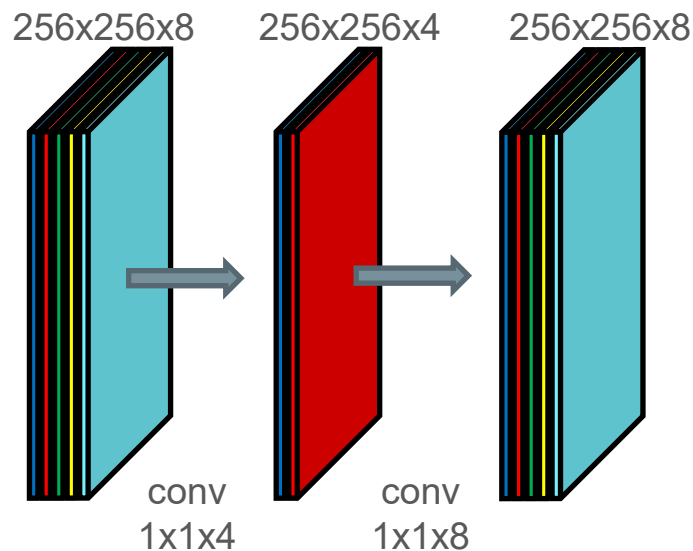


Bottleneck

Reduction of **spatial** dimension

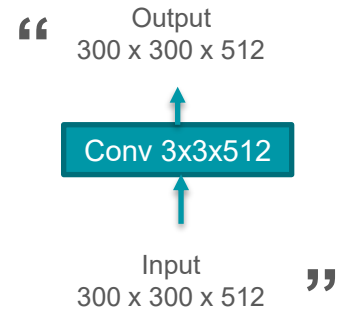
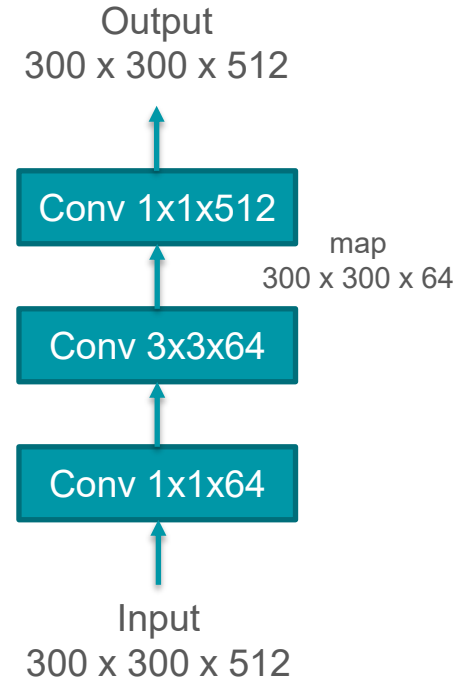


Reduction of **feature** dimension



Bottleneck layer

- for convolution neural networks
- to reduce the size of the input tensor in a convolutional layer
- lead to large saving in a computational cost
- preserves information with maximal relevance
- included in GoogLeNet (Inception V2 and V3) or DenseNet



Dilated convolution

- aims to aggregating the multi-scale contextual information

Standard convolution

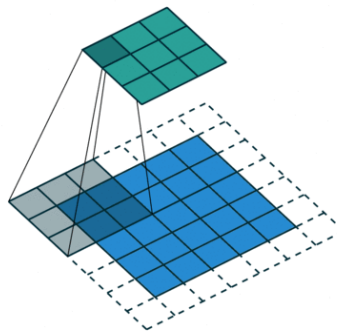
$$F(\mathbf{p}) * k(\mathbf{p}) = \sum_{\mathbf{p}=\mathbf{s}+\mathbf{t}} F(\mathbf{t})k(\mathbf{s})$$

Dilated convolution

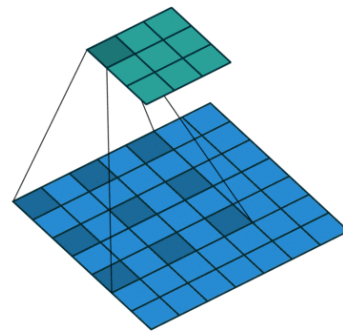
$$F(\mathbf{p}) *_{\delta} k(\mathbf{p}) = \sum_{\mathbf{p}=\mathbf{s}+\delta\mathbf{t}} F(\mathbf{t})k(\mathbf{s})$$

- no down-sampling
- the receptive field is larger without the need for the down and up-sampling
- analogously, the classic convolution with larger sparse kernel or pyramidal approach

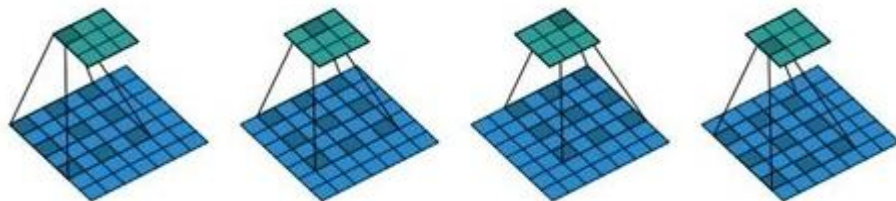
Dilation factor $\delta = 1$



$\delta = 2$



Dilated convolution with δ factor = 2



Up-sampling layers

The neural networks to generate images, it usually involves up-sampling from low resolution (from depth) to high resolution (output)

Un-pooling

- suitable for resampling (up) after a max pooling layer (non-linear)

Transposed convolution

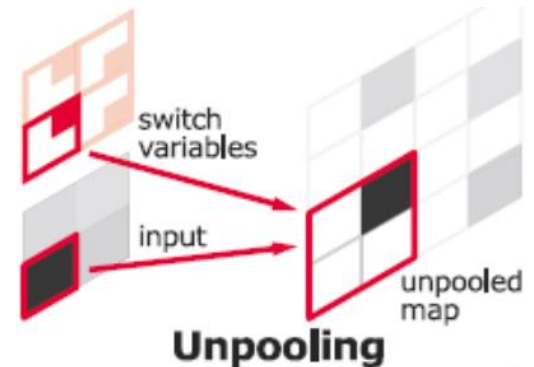
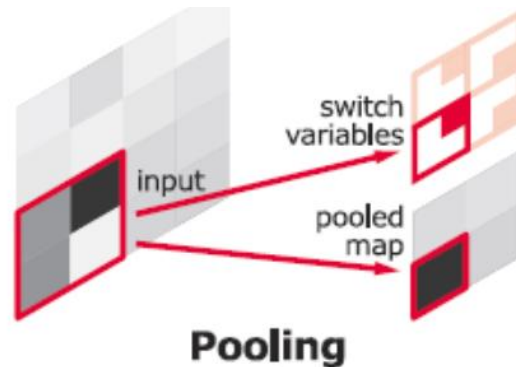
- This structure enables up-sampling matrices (features maps) from previous layers by convolution performed via matrix multiplication. Weights of convolution kernel are learned to achieved the best interpolation

Interpolation

- elementary interpolation method (bilinear)
- there is nothing that the network can learn about

Un-pooling layer (max)

- the task of this layer is up-sampling input (after pooling) – reverse pooling
- to perform un-pooling, we need to remember the position of each maximum activation value when doing max pooling
- in the un-pooling layer, the max value is inserted to the remembered position
- remains values are set to zeros
- there are no parameters for learning
- in the case, average pooling is performed by an interpolation method



Transposed convolution

- Sometimes wrongly called “deconvolution”
- It is using a computation of convolution via matrix multiplication...

$$G(\mathbf{p}) = F(\mathbf{p}) * k(\mathbf{p}) = \sum_{p=s+t} F(\mathbf{t})k(\mathbf{s}) \quad \longrightarrow \quad \text{vec}(\mathbf{G}) = \mathbf{C} \text{vec}(\mathbf{F})$$

... standard convolution with position vector $\mathbf{p}$

... where $\mathbf{G}$ is output matrix (convoluted matrix), $\mathbf{C}$ is **convolution matrix** and $\mathbf{F}$ is matrix of data

Note:

Then, inverse convolution can be derived to obtaining the matrix $\mathbf{F}$ from $\mathbf{G}$ and $\mathbf{C}$ as follows

$$\text{vec}(\mathbf{F}) = \mathbf{C}^{-1} \text{vec}(\mathbf{G})$$

*... **it is not possible**, because we cannot determine inverse convolution matrix $\mathbf{C}$.*

But we can use the convolution matrix to compute convolution (even stride-conv) and then use it for the up-sampling of $\mathbf{G}$ to the same size as matrix $\mathbf{F}$ (input to convolution).

Transposed convolution

$$\text{vec}(\mathbf{G}) = \mathbf{C} \text{vec}(\mathbf{F})$$

- Up-sampling method using **convolution matrix** with the specific size
- We can eliminate the impact of stride >1 in convolution layers
- Via matrix multiplication, we can effectively compute convolution with stride >1

Weights of kernel 3x3 ($K = 3$)

$$\begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \\ w_{3,1} & w_{3,2} & w_{3,3} \end{bmatrix}$$

Input image $M \times M$

After padding

$$\begin{bmatrix} x_{1,1} & \cdots & x_{1,M} \\ \vdots & \ddots & \vdots \\ x_{M,1} & \cdots & x_{M,M} \end{bmatrix}$$

$\mathbf{F}$

$$\Rightarrow \begin{bmatrix} x_{1,1} \\ x_{1,2} \\ \vdots \\ x_{1,M} \\ x_{2,1} \\ \vdots \\ x_{M,M} \end{bmatrix} \quad M^2$$

$\text{vec}(\mathbf{F})$

- To preserve the same size of output, we can apply a padding on input image $N \times N$
- We obtain a new size of image $M \times M$, where $M = N + (K - 1)$

Convolution matrix C

- rearranging of kernel to same size as input (after padding) image, thus $M \times M$

For standard convolution with stride = 1

$$\left(\frac{M}{\text{stride}}\right)^2 \begin{bmatrix} \text{for the first row of image} & \text{for the second row of image} & \text{for the third row of image} & \text{remaining rows} \\ \begin{matrix} w_{1,1} & w_{1,2} & w_{1,3} & 0_4 & \dots & 0_M \\ 0_1 & w_{1,1} & w_{1,2} & w_{1,3} & \dots & 0_M \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0_1 & 0_2 & 0_3 & \dots & \dots & 0_M \\ 0_1 & \dots & \dots & \dots & \dots & \dots \end{matrix} & \begin{matrix} w_{2,1} & w_{2,2} & w_{2,3} & 0_{M+4} & \dots & \dots & 0_{2M} \\ 0_{M+1} & w_{2,1} & w_{2,2} & w_{2,3} & 0_{M+5} & \dots & 0_{2M} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ w_{1,1} & w_{1,2} & w_{1,3} & 0_{M+4} & \dots & \dots & 0_{2M} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{matrix} & \begin{matrix} w_{3,1} & w_{3,2} & w_{3,3} & \dots & 0_{3M} & \dots & \dots & \dots & 0_{M^2} \\ 0_{2M+1} & w_{3,1} & w_{3,2} & w_{3,3} & \dots & \dots & \dots & \dots & 0_{M^2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0_{M^2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots & \dots & 0_{M^2} \end{matrix} & \begin{matrix} \dots & \dots & \dots & \dots & 0_{M^2-3} & w_{3,1} & w_{3,2} & w_{3,3} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{matrix} \end{bmatrix}$$

M^2

Each row of the matrix C represents a convolution calculation for single specific translation of kernel

Note: $M = N + (K - 1)$, N is a size of input image, K is size of kernel and M is size image after padding

Convolution matrix C

For **standard** convolution with **stride = 1**

$$\left(\frac{M}{1}\right)^2 \begin{bmatrix} \textcircled{w_{1,1}} & \textcircled{w_{1,2}} & \textcircled{w_{1,3}} & 0_4 & \cdots & 0_M & \textcircled{0_{M+1}} & w_{2,1} & w_{2,2} & w_{2,3} & 0_{M+4} & \cdots & \cdots & 0_{2M} & w_{3,1} & w_{3,2} & w_{3,3} & \cdots & 0_{3M} & \cdots & \cdots & \cdots & 0_{M^2} \\ 0_1 & w_{1,1} & w_{1,2} & w_{1,3} & \cdots & 0_M & 0_{M+1} & w_{2,1} & w_{2,2} & w_{2,3} & 0_{M+5} & \cdots & 0_{2M} & 0_{2M+1} & w_{3,1} & w_{3,2} & w_{3,3} & \cdots & \vdots & \vdots & \vdots & \vdots & 0_{M^2} \\ 0_1 & 0_2 & 0_3 & \vdots & \vdots & 0_M & \textcircled{w_{1,1}} & \textcircled{w_{1,2}} & \textcircled{w_{1,3}} & 0_{M+4} & \cdots & \cdots & 0_{2M} & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & 0_{M^2} \\ 0_1 & \vdots & 0_{M^2} \end{bmatrix}$$

M^2

For convolution with **stride = 2**

$$\left(\frac{M}{2}\right)^2 \begin{bmatrix} \text{for the first row of image} & \text{for the second row of image} & \text{for the third row of image} & \text{remaining rows} \\ \textcircled{w_{1,1}} & \textcircled{w_{1,2}} & \textcircled{w_{1,3}} & 0_4 & \cdots & 0_M & w_{2,1} & w_{2,2} & w_{2,3} & 0_{M+4} & \cdots & \cdots & 0_{2M} & w_{3,1} & w_{3,2} & w_{3,3} & \cdots & 0_{3M} & \cdots & \cdots & \cdots & 0_{M^2} \\ 0_1 & 0_2 & w_{1,1} & w_{1,2} & \cdots & 0_M & 0_{M+1} & 0_{M+1} & w_{2,1} & w_{2,2} & w_{2,3} & \cdots & 0_{2M} & 0_{2M+1} & 0_{2M+1} & w_{3,1} & w_{3,2} & \cdots & \vdots & \vdots & \vdots & \vdots & 0_{M^2} \\ \vdots & 0_{M^2} \\ 0_1 & 0_2 & 0_3 & \vdots & \vdots & 0_M & \vdots & \vdots & \vdots & \vdots & 0_{M+4} & \cdots & 0_{2M} & \textcircled{w_{1,1}} & \textcircled{w_{1,2}} & \textcircled{w_{1,3}} & \cdots & \vdots & \vdots & \vdots & \vdots & 0_{M^2} \\ 0_1 & \vdots & 0_{M^2} \end{bmatrix}$$

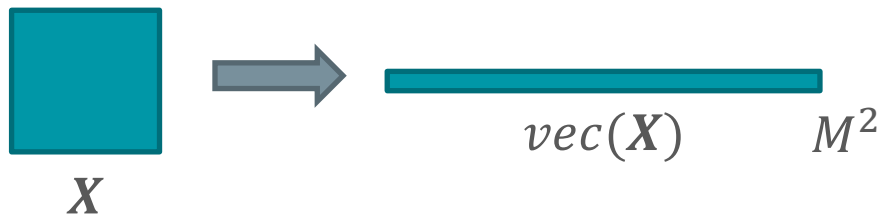
M^2

If stride = 2, the column of C for the second row (and every other row) of the image is equal to 0

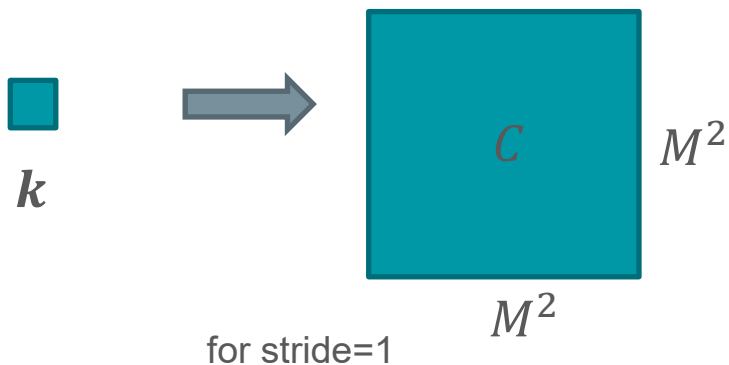
Note: $M = N + (K - 1)$, N is a size of input image, K is size of kernel and M is size image **after padding**

Transposed convolution

1. Vectorization of activation map (image) sized $M \times M$



2. Creation of sparse convolution matrix $\mathbf{C}$ from kernel



Convolution matrix for conv with stride=2

$$\left(\frac{M}{stride}\right)^2 \quad \text{[teal rectangle]} \quad M^2$$

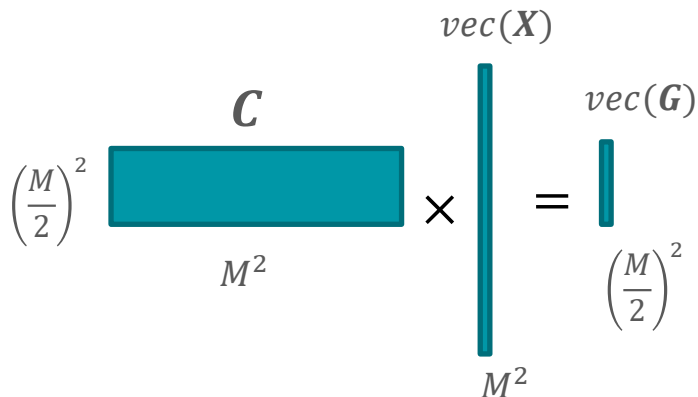
Transposed convolution

Then, we easily can up-sample activation maps into the original size (before convolution layer with stride > 1, which caused its sub-sampling).

We need only the size of the stride and size of sub-sampled map. We create an initial convolution matrix $\mathbf{K}$ with the same size as matrix $\mathbf{C}$, whose values will be learned.

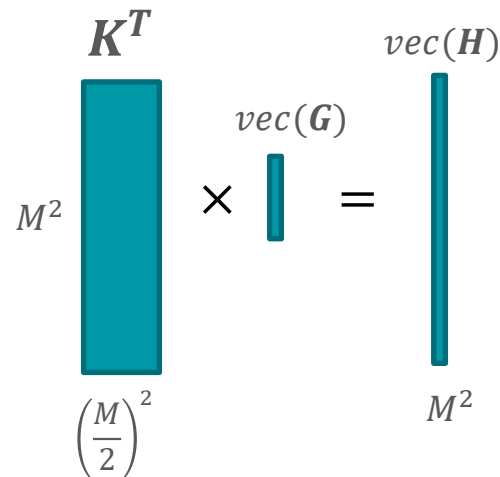
Convolution with stride=2

$$\text{vec}(\mathbf{G}) = \mathbf{C} \text{vec}(\mathbf{X})$$



Transposed convolution with stride=2

$$\text{vec}(\mathbf{H}) = \mathbf{K}^T \text{vec}(\mathbf{B})$$



Attention module

- The main idea is, in the first stage, the network pays attention to the global (overall) context
- Then, it can focus on details in the full resolution
- For tasks which require a global perceiving, not point-based approach

- Fully connected–based networks
- Residuals blocks
- The decoder uses an activation map from each level

- **Attention map** – a weight matrix that defines the area of interest (localization/feature) in the activation map.

- Provides contextual information about the problem
- Useful in recurrent nets (eliminate vanishing problem)
- For segmentation task – spatial information
- Basic unit of special network called “Transformers”

- high computational complexity (fully connected)

